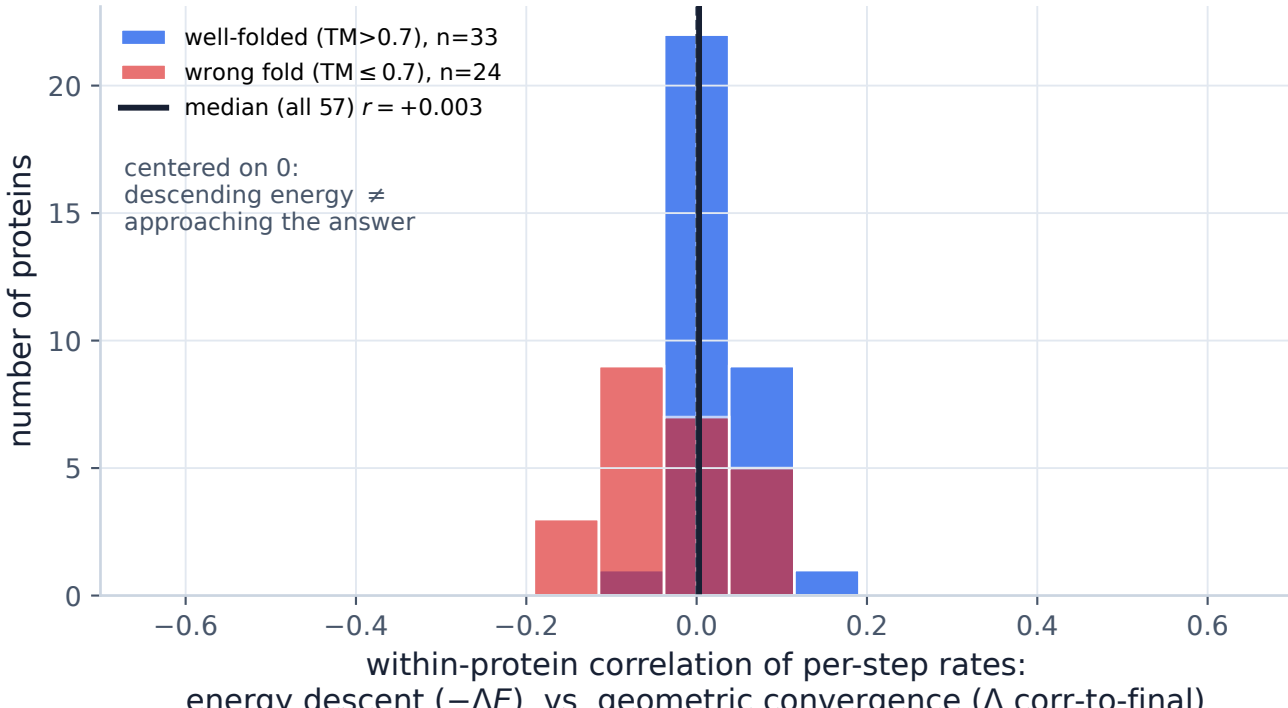Energy descent is decoupled from convergence to the answer